

# Yasmine Gharsallah

+33 754 863 850 | [yasmine.gharsallah@eurecom.fr](mailto:yasmine.gharsallah@eurecom.fr)

 [yasmine-gharsallah](#) |  [Yasmine-Gharsallah](#) | Antibes, France

## SUMMARY

Final-year ICT and Cybersecurity Engineering student completing a double degree in Security of Computer Systems and Communications at EURECOM. Strong foundation in secure systems, cloud infrastructures, and network technologies, with hands-on experience across cryptography, hardware security, and applied AI. Seeking a 5 to 6-month end-of-studies internship starting from February 2027 in Security of Computer Systems and Communications .

## EDUCATION

- Eurecom: Graduate School and Research Center in Digital Science**  September 2025 – Present  
*Post-Master's: Security in Computer Systems and Communications* Biot, France  
**Relevant Coursework:** Hardware Security, System and Network Security, Secure Communications, Imaging Security, Digital Forensics, Wireless Security.
- The Higher School of Communications of Tunis (SUP'COM)**  September 2023 – Present  
*ICT Engineering Student* Tunis, Tunisia  
**Relevant Coursework:** Advanced Malware Reverse Engineering, Cryptography, Hardware Security, FMCW Radar and AI, Antenna Theory, Probability, Algorithmic Foundations, Artificial Intelligence, Big Data, Cloud Fundamentals, Optimization.
- Preparatory Institute for Engineering Studies of Sfax**  September 2021 – June 2023  
*Pre-engineering Studies* Sfax, Tunisia
  - Ranked 179/1780 in the National Entrance Exams for Engineering Training Cycles – MP track (Mathematics and Physics)

## EXPERIENCE

- Technical and Customer Service Center, Tunisie Telecom** 1st July 2024 – 3rd August 2024  
*Summer Internship* Ariana, Tunisia
  - Gained hands-on experience in subscriber network architecture (RLA), copper and fibre optic technologies (FTTC/FTTH), fault management processes, and telecommunication line configurations. 

## PROJECTS

- Fairness and Differential Privacy in Federated Learning on Knowledge Graphs** March 2025 – Present  
*Tools: Python, PyTorch, NetworkX, Louvain, FedAvg, NELL-995*
  - Studying fairness on Knowledge Graphs in federated learning settings, analysing bias propagation through entity relationships and semantic embeddings under non-IID data distributions, and exploring Differential Privacy mechanisms to enhance fairness while preserving privacy.
- Investigation of Phone Duration Impacts in Voice Anonymisation** October 2024 – February 2025  
*Tools: Python, librosa, WORLD Vocoder, Montreal Forced Aligner (MFA), PyTorch (BiGRU)*
  - Developed speech processing pipelines to extract and analyse phone-level durations from original and anonymised speech.
  - Evaluated anonymisation robustness using three re-identification attacks (metric-based, duration embedding, and semi-informed filter-bank) under the VoicePrivacy Challenge protocol, measuring Equal Error Rate (EER) on LibriSpeech subsets.
- Traffic Monitoring: mmWave RADAR and Camera Detection, Classification, and High-Level Fusion** April – May, 2025  
*Tools: YOLO v8, KNN, DBSCAN, Python*
  - Built a traffic monitoring system using the Open Urban mmWave Radar and Camera Vehicle Classification Dataset. Applied feature extraction, clustering with DBSCAN, classification with KNN, and object detection with YOLO. Implemented high-level sensor fusion to enhance detection accuracy and system reliability.
- Secure Cloud-Based Document Management System for Confidential Information** January – May, 2025  
*Tools: Angular, Spring Boot, AWS Cognito, AWS S3, AWS RDS, AWS Lambda, AWS IAM, PostgreSQL, REST APIs*
  - Developed a web-based platform for enterprises to securely manage, share, and track confidential documents, with fine-grained access control, multi-factor authentication, and a serverless cloud architecture.
- DES Implementation and CPA Side-Channel Attack in VHDL** January – February, 2025  
*Tools: VHDL, Xilinx Vivado, Xilinx Artix-7 FPGA (XC7A35T)*
  - Implemented and analysed a DES algorithm in VHDL using hardware simulation, and exploited internal signal transitions to emulate a Correlation Power Analysis (CPA) attack.

## SKILLS

- Programming Languages:** Python, SQL, SQLITE, C, C++, Matlab
- AI & ML Frameworks:** PyTorch, NetworkX, librosa, scikit-learn, YOLO
- Cryptography & Secure Communications:** RSA, Diffie-Hellman Key Exchange, Elliptic Curve Cryptography (ECC), Symmetric/Asymmetric Encryption, Hash Functions, PKI, TLS/SSL
- Web and Mobile Development:** Angular, Spring Boot, Flutter, Firebase, HTML, CSS, JavaScript
- Database Systems:** Oracle, PostgreSQL
- Cloud Platforms:** Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP)
- Other Tools & Technologies:** Keysight ADS, Cisco Packet Tracer, Wireshark, Linux

## CERTIFICATIONS

- SOC level1 (Try Hack me)** In Progress
- DevSecOps (Try Hack me)** In Progress
- Hashgraph Developer Course** December 2024
- Introduction to Cybersecurity Learning Path (Try Hack Me)** August 2024
- Introduction to Cybersecurity (CISCO)** July 2024
- CCNAv7: Introduction to Networks (CISCO)** September 2024

## ADDITIONAL INFORMATION

**Languages:** English (Professional, TOEIC: 915/990), French (Fluent), Arabic (Native), Spanish (Beginner, A1)